

Data Communications and Networking Fourth Edition

Forouzan

Unit -1

Chapter 3

Data and Signals

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Note

To be transmitted, data must be transformed to electromagnetic signals.

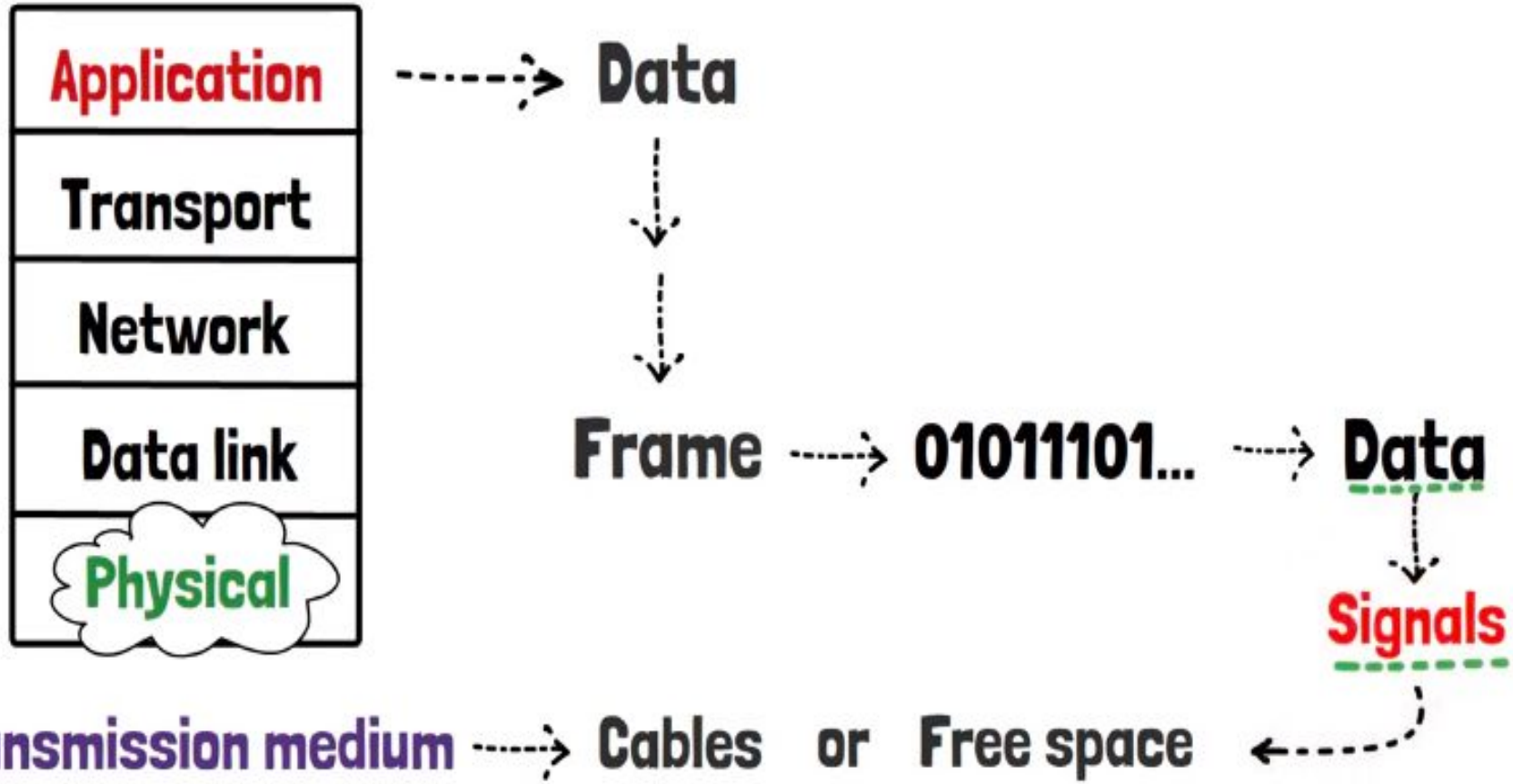
3-1 ANALOG AND DIGITAL

Data can be analog or digital. The term analog data refers to information that is continuous; digital data refers to information that has discrete states. Analog data take on continuous values. Digital data take on discrete values.

Topics discussed in this section:

- **Analog and Digital Data**
- **Analog and Digital Signals**

TCP/IP protocol suite



Data

Analog

{ Continuous }



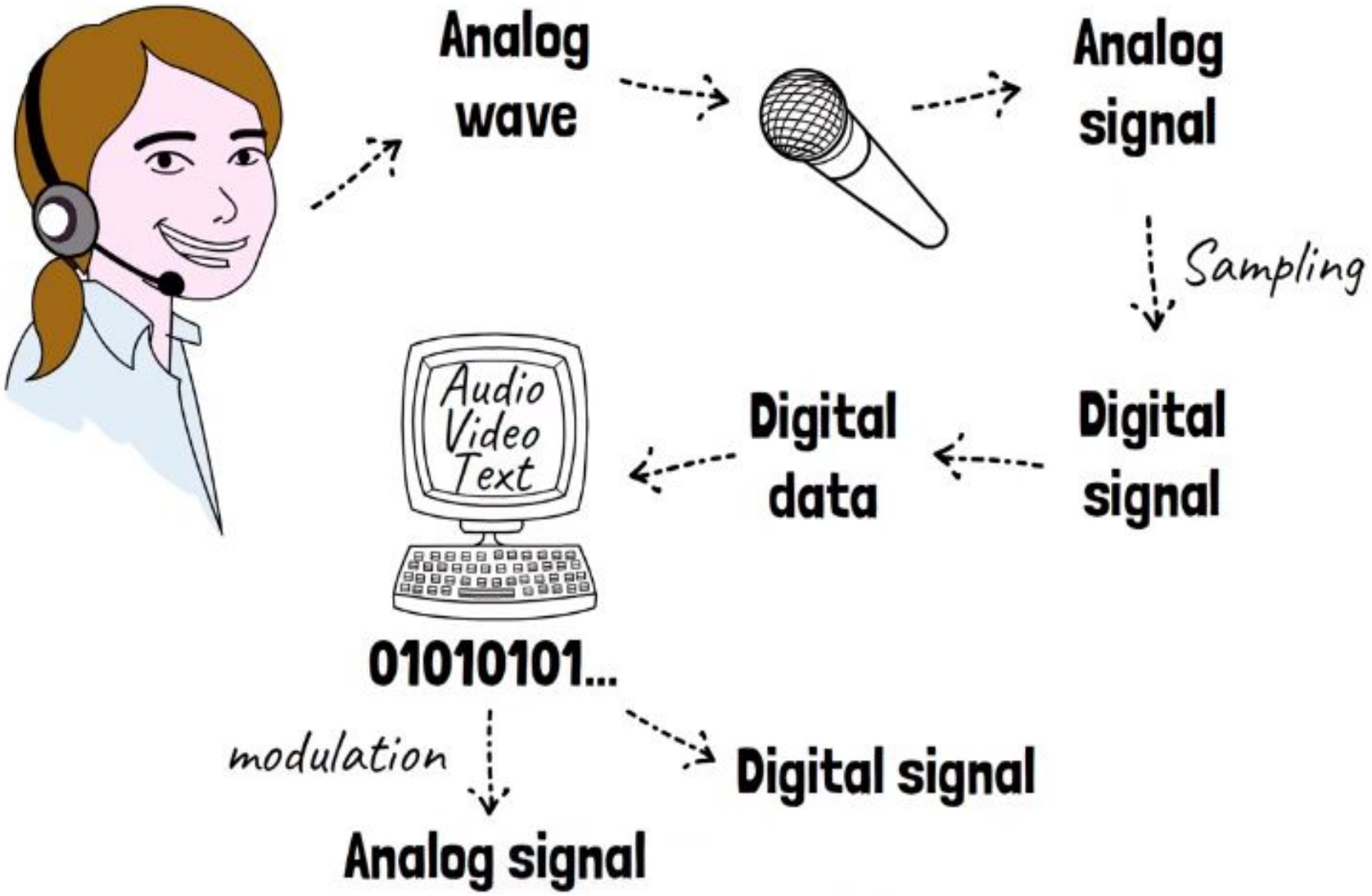
10:10 to 10:11

Digital

{ discrete }



10:11

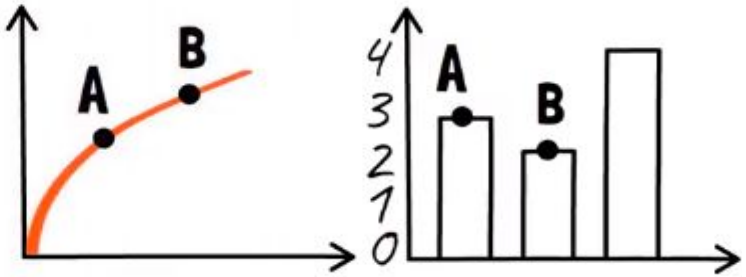

Analog wave
Analog signal
Sampling
Digital signal
Digital data
Audio
Video
Text
01010101...
modulation
Digital signal
Analog signal

Signals

Analog Digital

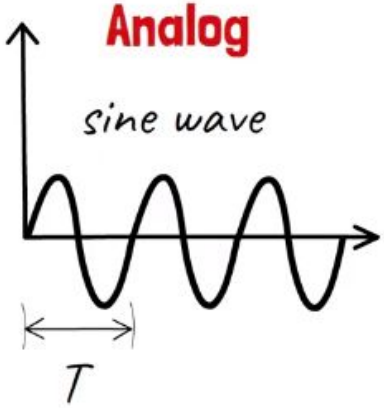
Periodic

Non-periodic



Periodic Signal

Analog



Non-Periodic Signal

Digital



Analog and Digital Data

- Data can be analog or digital.
- Analog data are continuous and take continuous values.
- Digital data have discrete states and take discrete values.

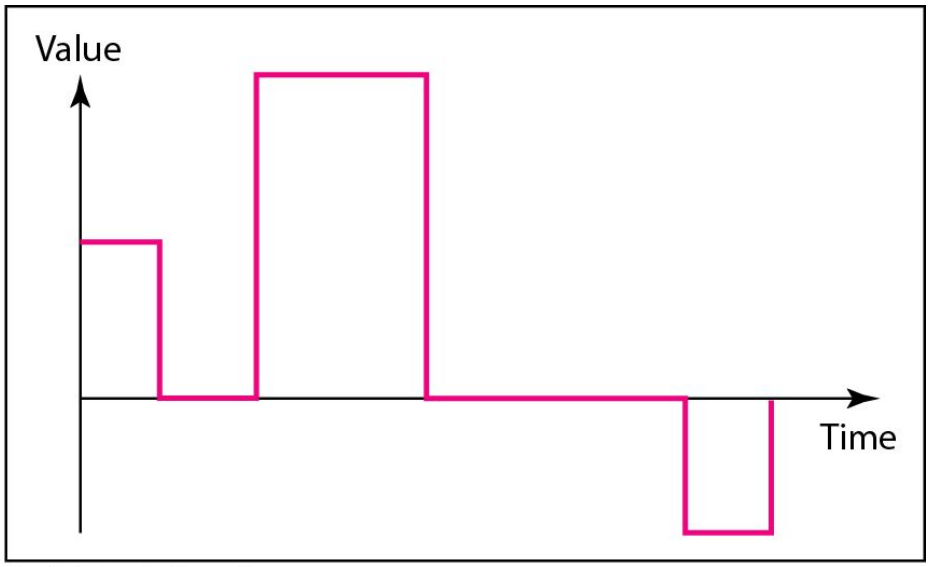
Analog and Digital Signals

- Signals can be analog or digital.
- Analog signals can have an infinite number of values in a range.
- Digital signals can have only a limited number of values.

Figure 3.1 *Comparison of analog and digital signals*



a. Analog signal



b. Digital signal

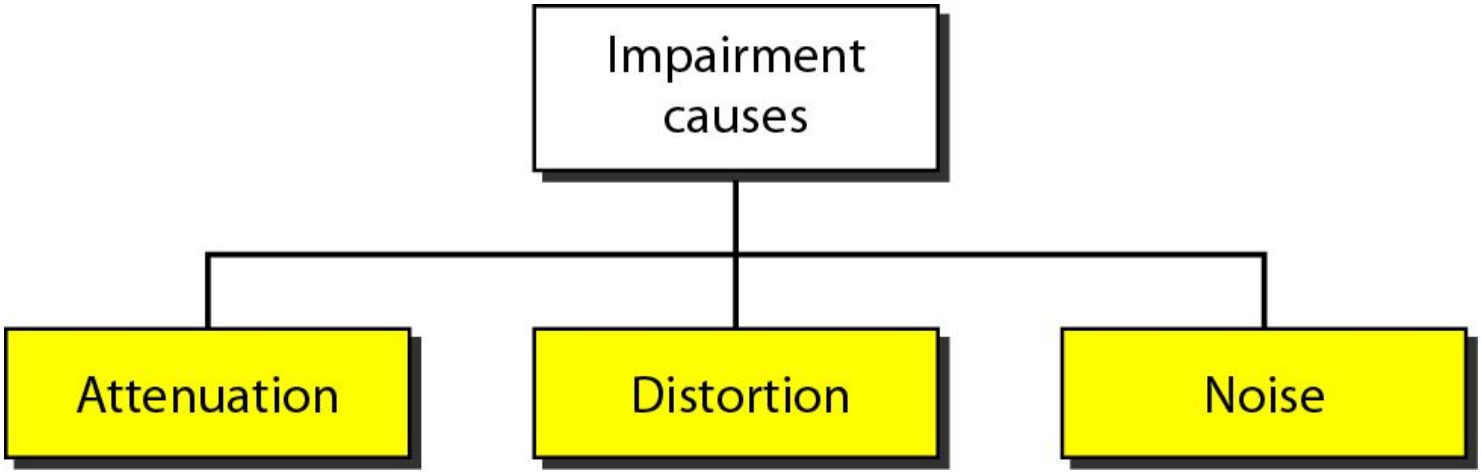
3-4 TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

Topics discussed in this section:

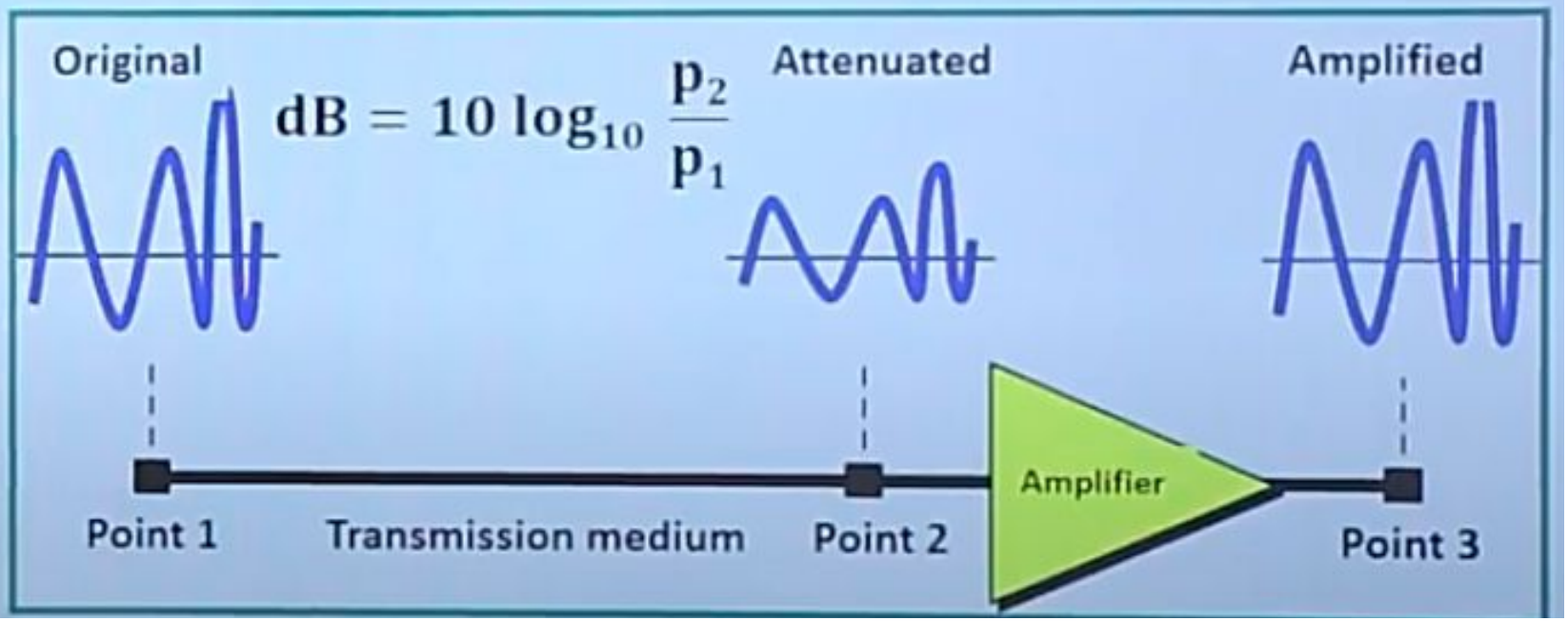
- **Attenuation**
- **Distortion**
- **Noise**

Figure 3.25 *Causes of impairment*



Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.

Decibel (dB) will be calculated by taking ratio of power of signal at two different points. Let P1 and P2 are the powers at point 1 and point 2.



1. Attenuation

- Loss of energy
- Compensate: Amplifiers
- Unit to measure strength: Decibel(dB)
- Decibel
  -Positive: amplified
  -Negative: Attenuated

- **FORMULA :**

$$dB = 10 \log_{10}(P_2 / P_1)$$

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1.A Signal travels through a transmission medium and its power is reduced to one half. Calculate the attenuation



P_1 $P_2 = 0.5\ P_1$

$dB = 10 \log_{10}(P_2 / P_1)$

$= 10 \log_{10}(0.5\ P_1 / P_1)$

$= 10 \log_{10}(0.5)$

$= 10*(-0.3010)$

$= -3.01$ dB

$= -3$ dB approx. loss

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2.A Signal travels through a amplifier and its power is increased 20 times. Calculate the power gained



P_1 $P_2 = 20\ P_1$

$dB = 10 \log_{10}(P_2 / P_1)$

$= 10 \log_{10}(20\ P_1 / P_1)$

$= 10 \log_{10}(20)$

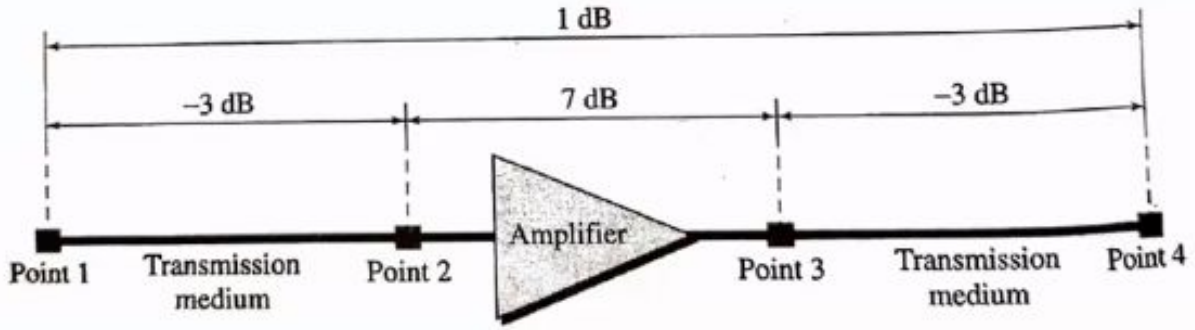
$= 10*(1.301)$

$= 13.01$ dB

$= 13$ dB approx. gain

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3.(Refer fig) **A signal travels from point 1 to point 4. Signal is attenuated by the time it reaches point 2 (-3dB) . Between points 2 and 3 signal is amplified (7dB) . Again between points 3 and 4 signal is attenuated (-3dB).What is the resultant decibel value.**



dB= -3 +7 -3 =+1

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Decibel used to measure power in milliwatts

$$dB_m = 10 \log_{10} P_m$$

P_m :power in milliwatts

1. Calculate the power of a signal if its dB_m is -30

$$dB_m = 10 \log_{10} P_m$$

$$-30 = 10 \log_{10} P_m$$

$$10 \log_{10} P_m = -30$$

$$\log_{10} P_m = -30/10$$

$$\log_{10} P_m = -3$$

$$P_m = 10^{-3} \text{ m W}$$

$\log_a x = y$

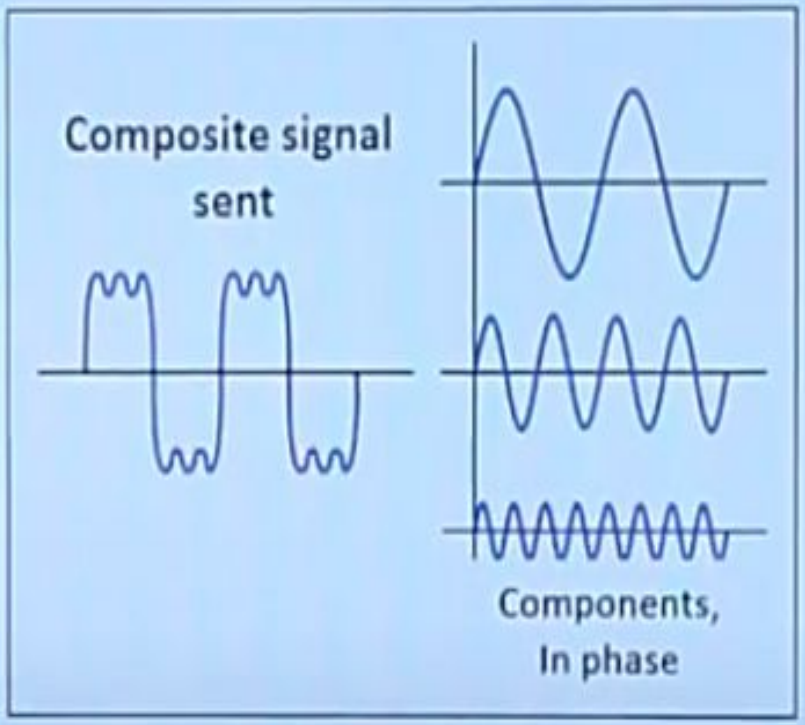
$x = a^y$

$\log_{10} P_m = -3$

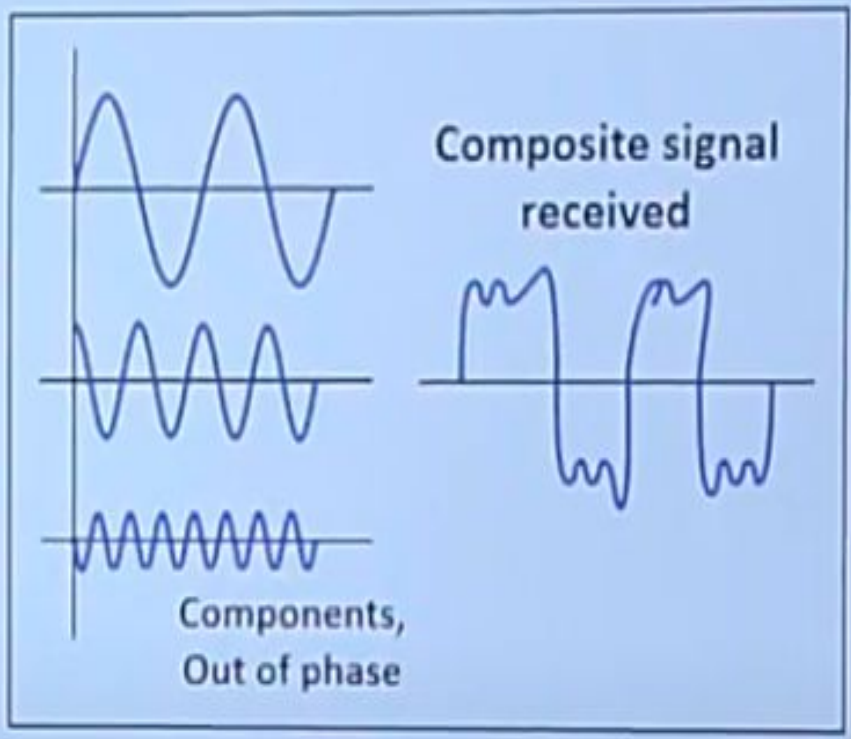
$P_m = 10^{-3}$

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Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies and having variability in their propagation speed.

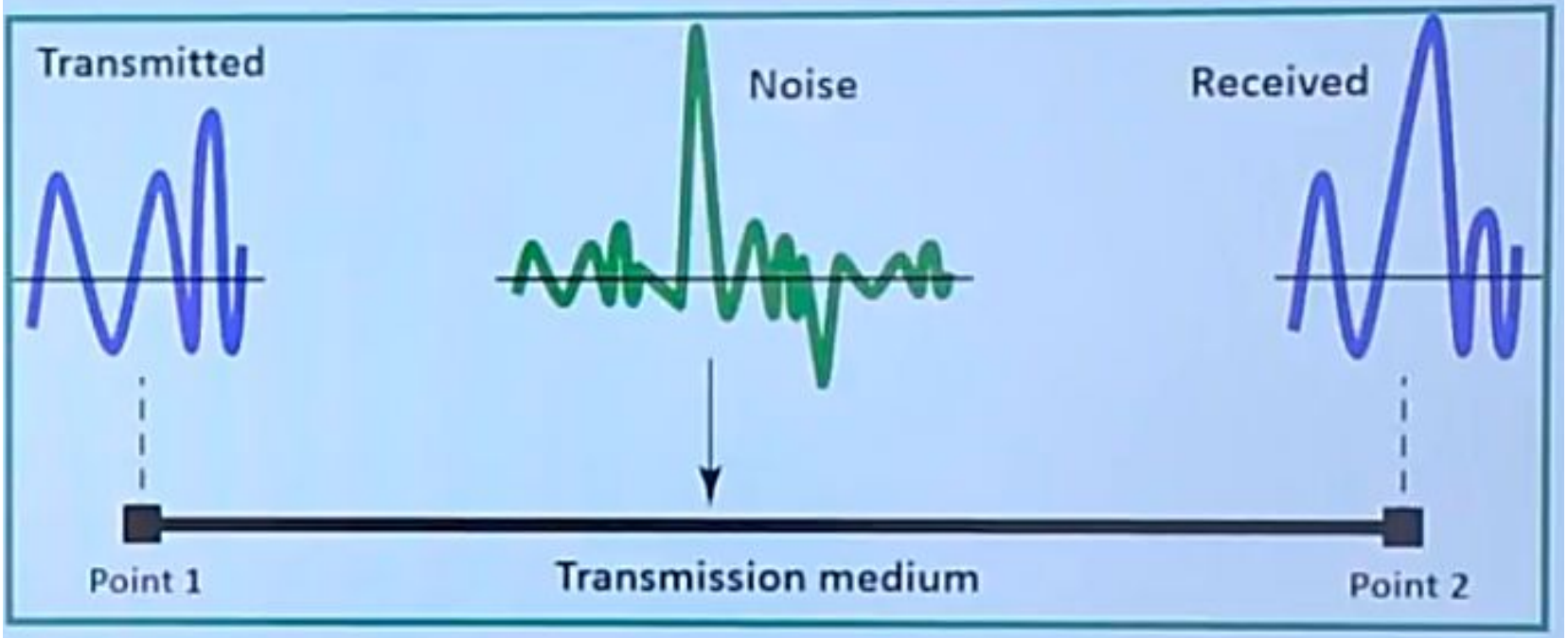


At the sender



At the receiver

Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.

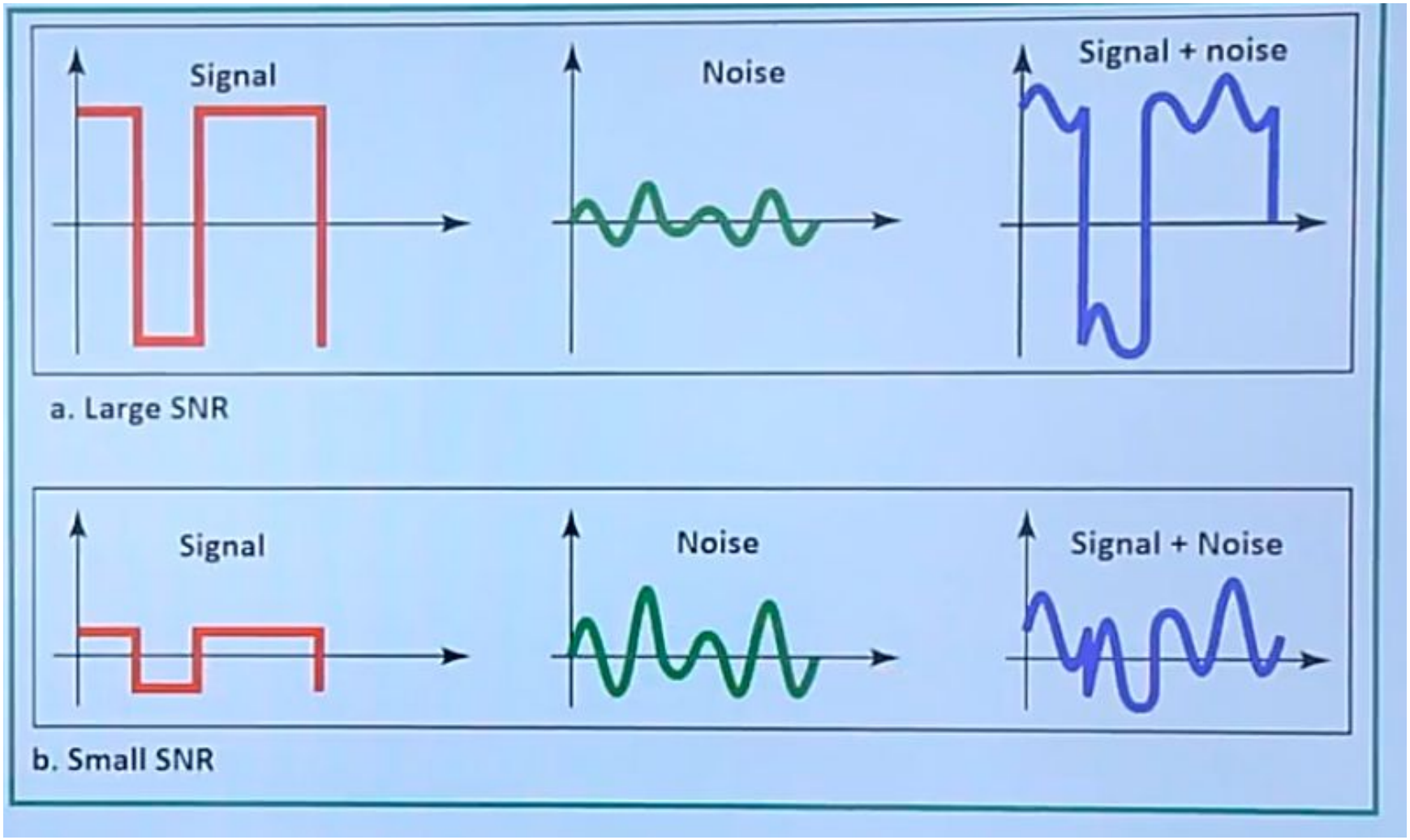


Signal to noise ratio is defined as,

SNR = (Average Signal Power) / (Average Noise Power)

and

$$\mathbf{SNR_{dB} = 10 \log_{10} SNR}$$


Signal
Noise
Signal + noise
a. Large SNR
Signal
Noise
Signal + Noise
b. Small SNR

Average Power of a signal is 10 mW and Average power of noise is 1μW. What are the values of SNR and SNR_{dB}

$$\textbf{SNR} = \frac{\textbf{Average Signal Power}}{\textbf{Average noise Power}}$$

$$SNR = \frac{10^{-2}}{10^{-6}}$$

$$= 10^{-2} * 10^{6}$$

$$= 10^{4}$$

$$= 10000$$

$1\ W = 10^{3}\ mW$

$1\ mW = 10^{-3}\ W$

$10\ mW = 10^{-2}\ W$

$1\ W = 10^{6}\ \mu W$

$1\ \mu W = 10^{-6}\ W$

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Power of a signal is 10 mW and power of noise is 1μW. What are the values of SNR and SNR_{dB}

$$SNR = \frac{\text{Average Signal Power}}{\text{Average noise Power}}$$

$$SNR = \frac{10^{-2}}{10^{-6}}$$

$$= 10^{-2} * 10^{6}$$

$$= 10^{4}$$

$$= 10000$$

$$SNR_{dB} = 10 \log_{10} SNR$$

$$SNR_{dB} = 10 \log_{10} SNR$$

$$= 10 \log_{10} (10000)$$

$$= 10 * 4$$

$$= 40$$

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Topic

- **DATA RATE LIMITS**
  - -Nyquist Bit rate
  - -Shannon capacity

Textbook referred: Data Communication and networking (Forouzan)

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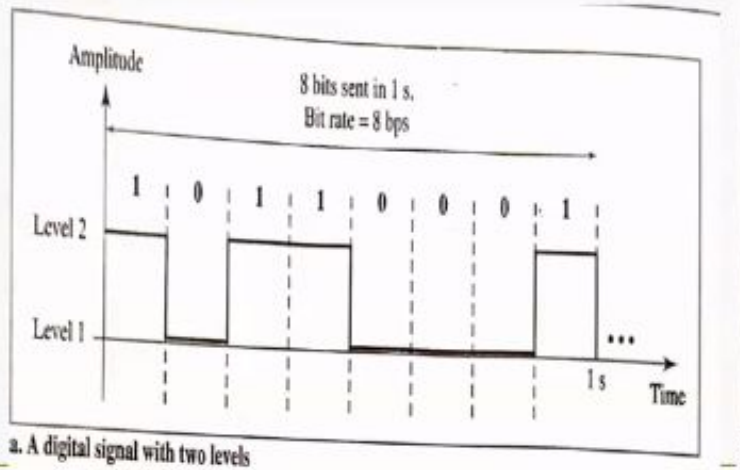
Data rate refers to the speed of data transfer through a channel. It is generally computed in bits per second (bps). Higher data rates are expressed as Kbps ("Kilo" bits per second, i.e.1000 bps), Mbps ("Mega" bits per second, i.e.1000 Kbps), Gbps ("Giga" bits per second, i.e. 1000 Mbps) and Tbps ("Tera" bits per second, i.e. 1000 Gbps).

One of the main objectives of data communications is to increase the data rate. There are three factors that determine the data rate of a channel:

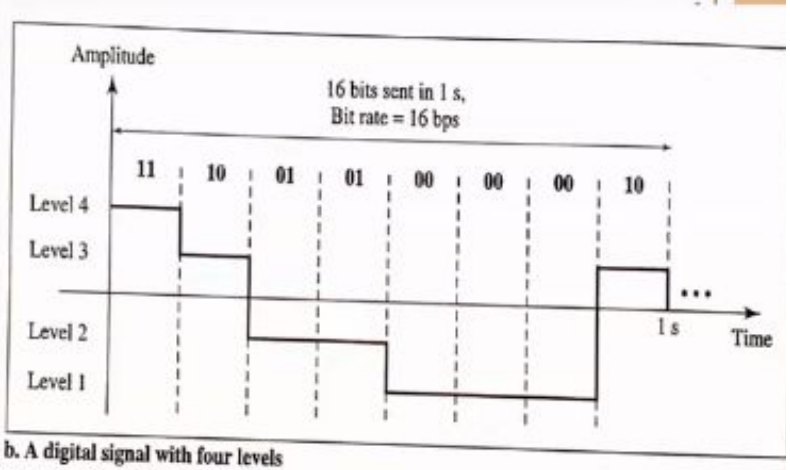
DATA RATE LIMITS

DATA RATE FACTORS :

- Bandwidth
- Level of signals
- Channel Quality



a. A digital signal with two levels



b. A digital signal with four levels

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DATA RATE LIMITS

DATA RATE FACTORS :

- Bandwidth
- Level of signals
- Channel Quality

Two formulas to calculate data rate:

- Nyquist Bit Rate : Noiseless Channel
- Shannon Capacity : Noisy Channel

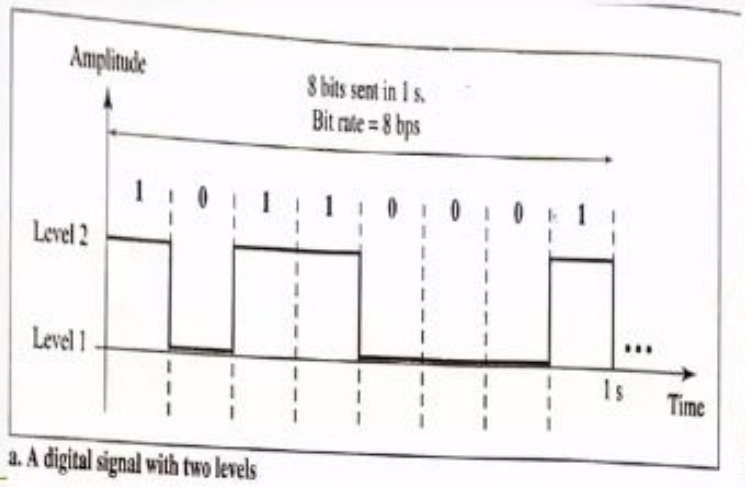
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DATA RATE LIMITS

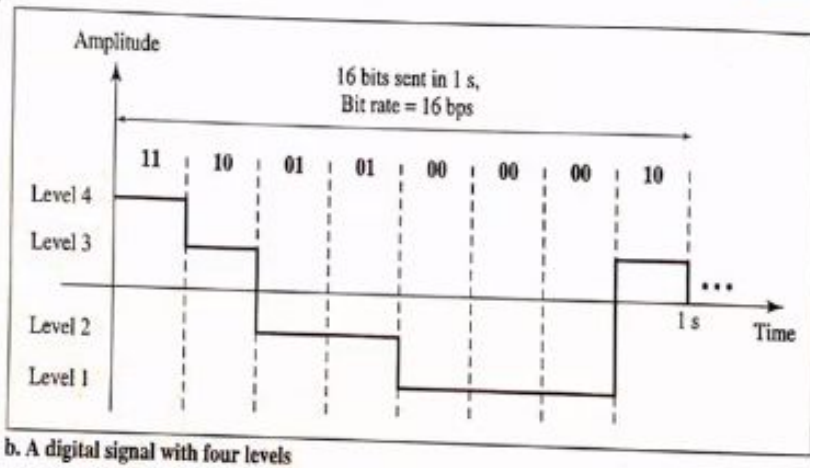
Nyquist Bit Rate : Noiseless Channel

- **Maximum bit rate**

$$\text{Bit Rate} = 2 * \text{Bandwidth} * \log_2 L$$



a. A digital signal with two levels



b. A digital signal with four levels

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Nyquist Bit Rate : Noiseless Channel

1. A noiseless channel has a bandwidth of 4000 Hz and is transmitting a signal with two signal Levels. Calculate the maximum bit rate.

$$\text{Bit Rate} = 2 * \text{Bandwidth} * \log_2 L$$

$$\text{Bit Rate} = 2 * 4000 * \log_2 2$$

$$= 2 * 4000 * 1$$

$$= 8000 \text{ bps}$$

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Nyquist Bit Rate : Noiseless Channel

2. A noiseless channel has a bandwidth of 4000 Hz and is transmitting a signal with four signal Levels. Calculate the maximum bit rate.

Bit Rate = 2 * Bandwidth * $\log_2 L$

Bit Rate $= 2 * 4000 * \log_2 4$

$= 2 * 4000 * 2$

$= 16{,}000$ bps

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Nyquist Bit Rate : Noiseless Channel

3. Consider a noiseless channel with a bandwidth of 20 KHz. We need to send 280 kbps over a channel. How many signal levels are required?

$$\text{Bit Rate} = 2 * \text{Bandwidth} * \log_2 L$$

$280 = 2 * 20 * \log_2 L$

$280/40 = \log_2 L$

$7 = \log_2 L$

$\log_2 L = 7$

$L = 2^7$

$L = 128$

$\log_a x = y$

$x = a^y$

$\log_2 L = 7$

$L = 2^7$

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Shannon Capacity

- **Shannon Capacity : Noisy Channel**

$$\text{Capacity} = \text{Bandwidth} * \log_2(1+\text{SNR})$$

$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average noise Power}}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

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Shannon Capacity : Noisy Channel

1. Consider a extremely noisy channel in which signal to noise ratio is almost zero. Calculate the capacity of the channel.

$$\text{Capacity} = \text{Bandwidth} * \log_2(1+\text{SNR})$$

$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average noise Power}}$$

$$\begin{aligned}\text{Capacity} &= B * \log_2(1+\text{SNR}) \\ &= B * \log_2(1+0) \\ &= B * \log_2(1) \\ &= B*0 \\ &= 0\end{aligned}$$

Capacity zero regardless of bandwidth

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Shannon Capacity : Noisy Channel

2. Calculate the highest bit rate(capacity of a channel) if the bandwidth is 3000Hz and signal to noise ratio(SNR) is 3162.

Capacity =Bandwidth * $\log_2(1+SNR)$

Capacity =B * $\log_2(1+SNR)$

= 3000 * $\log_2(1+3162)$

= 3000 * $\log_2(3163)$

=3000*11.627

= 34881 bps

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Shannon Capacity : Noisy Channel

3. Assume that SNR_{dB} =36 and bandwidth of channel is 2MHz. Calculate channel capacity.

$$SNR_{dB} = 10 \log_{10} SNR$$

$$\text{Capacity} = \text{Bandwidth} * \log_2(1+SNR)$$

$SNR_{dB} = 10 \log_{10} SNR$

$SNR_{dB}/10 = \log_{10} SNR$

$\log_{10} SNR = SNR_{dB}/10$

$SNR = 10^{SNR_{dB}/10}$

$SNR = 10^{36/10}$

$SNR = 10^{3.6}$

$SNR = 3981$

$\log_a x = y$

$x = a^y$

$\log_{10} SNR = SNR_{dB}/10$

$SNR = 10^{SNR_{dB}/10}$

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Shannon Capacity : Noisy Channel

3. Assume that SNR_{dB} =36 and bandwidth of channel is 2MHz. Calculate channel capacity.

Capacity =Bandwidth * $\log_2(1+SNR)$

$SNR_{dB} = 10 \log_{10} SNR$

$SNR_{dB}/10 = \log_{10} SNR$

$\log_{10} SNR = SNR_{dB}/10$

$SNR = 10^{SNR_{dB}/10}$

$SNR = 10^{36/10}$

$SNR = 10^{3.6}$

$SNR = 3981$

Capacity $= B * \log_2(1+SNR)$

$= 2 * \log_2(1+3981)$

$= 2 * \log_2(3982)$

$= 2 * 11.959$

$= 23.91$

$= 24$ Mbps approx.

$= 24 * 10^6$ bps

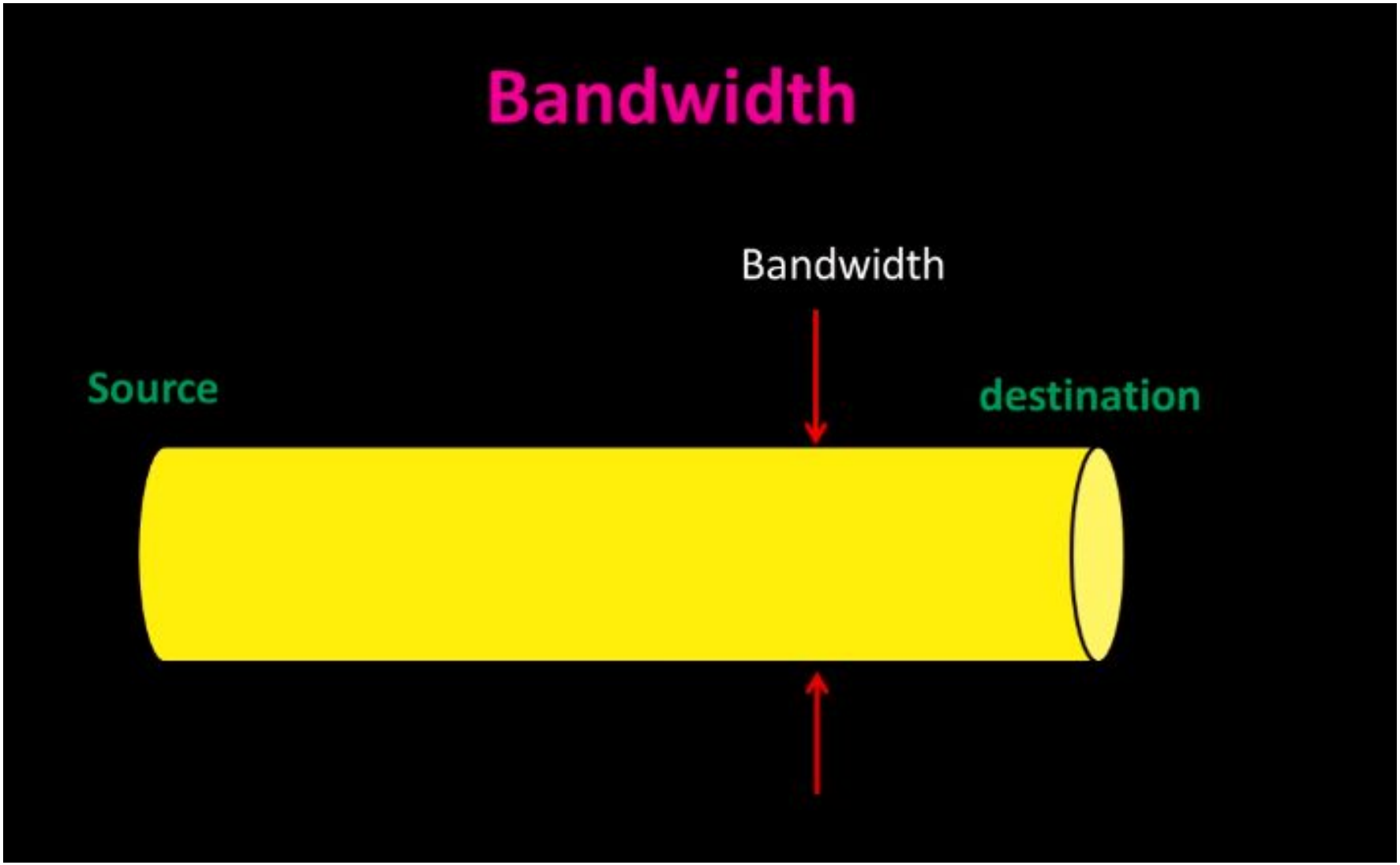
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Performance

Bandwidth

- Bandwidth is **the maximum amount of information transmitted** over a connection in a specific amount of time.
- Bandwidth is indicating the maximum network's capacity of communications link.
- Represented in bit/sec or byte/sec.

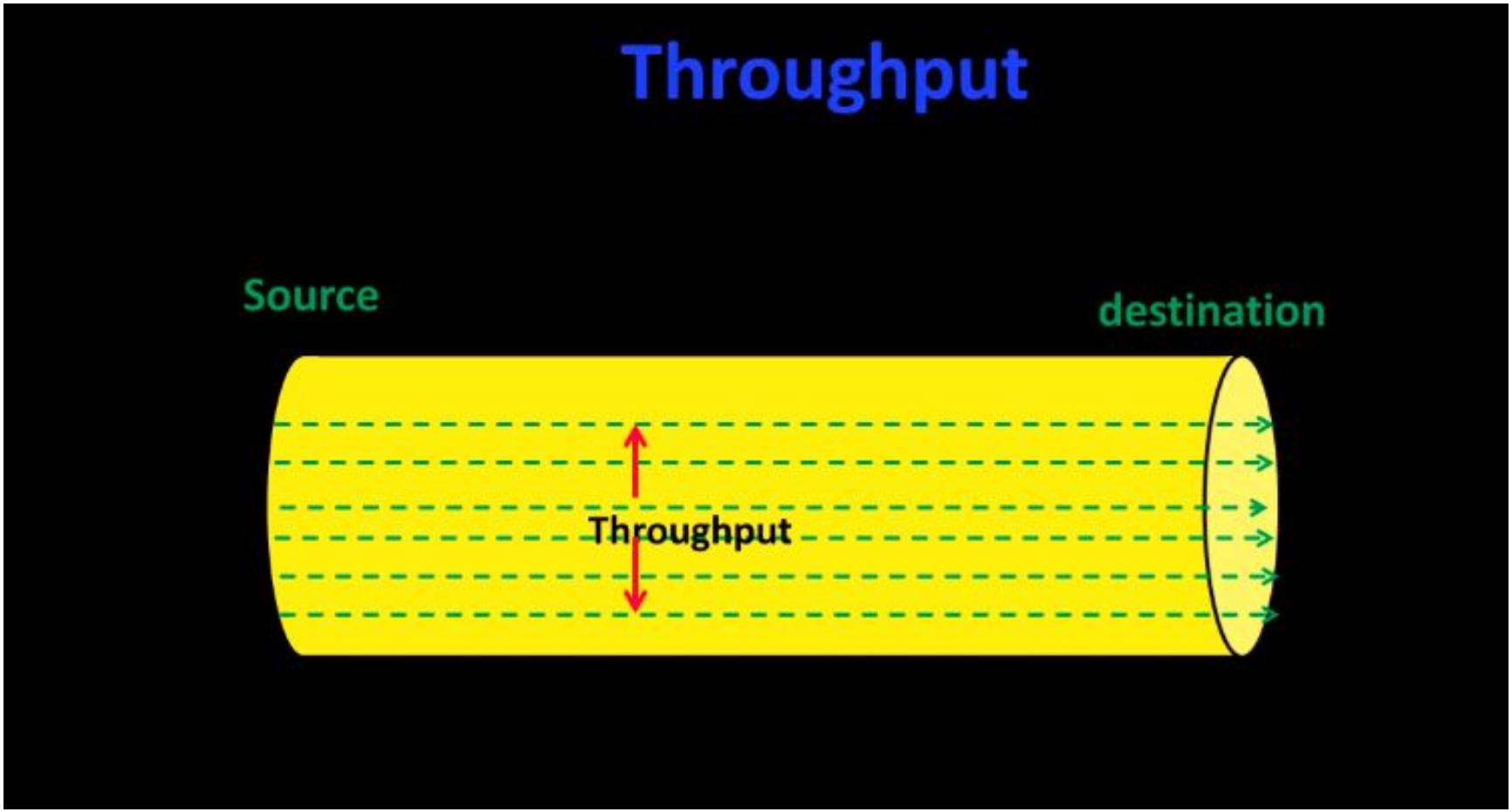
  Eg:bps, kbps, Mbps, Gbps.
- The more bandwidth, the more data it can send and receive .


Bandwidth
Bandwidth
Source
destination

Throughput

- Throughput is **the actual amount of data** passing through connection.
- Throughput means how many data packets reach at destination.
- Throughput capacity is measured in bit/sec or byte/sec.

  Eg:bps, kbps, Mbps, Gbps.


Throughput
Source
destination
Throughput

Example 1.18

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\text{Throughput} = (12{,}000 \times 10{,}000) / 60 = 2 \text{ Mbps}$$

1.12.3 Latency (Delay)

• The latency defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

Latency = propagation time + transmission time + queuing time + processing delay

1) Propagation Time

- Propagation time is defined as the time required for a bit to travel from source to destination.

Propagation time is given by

Propagation time = Distance / (Propagation Speed)

- Propagation speed of electromagnetic signals depends on
  - → medium and
  - → frequency of the signal.

Example 1.19

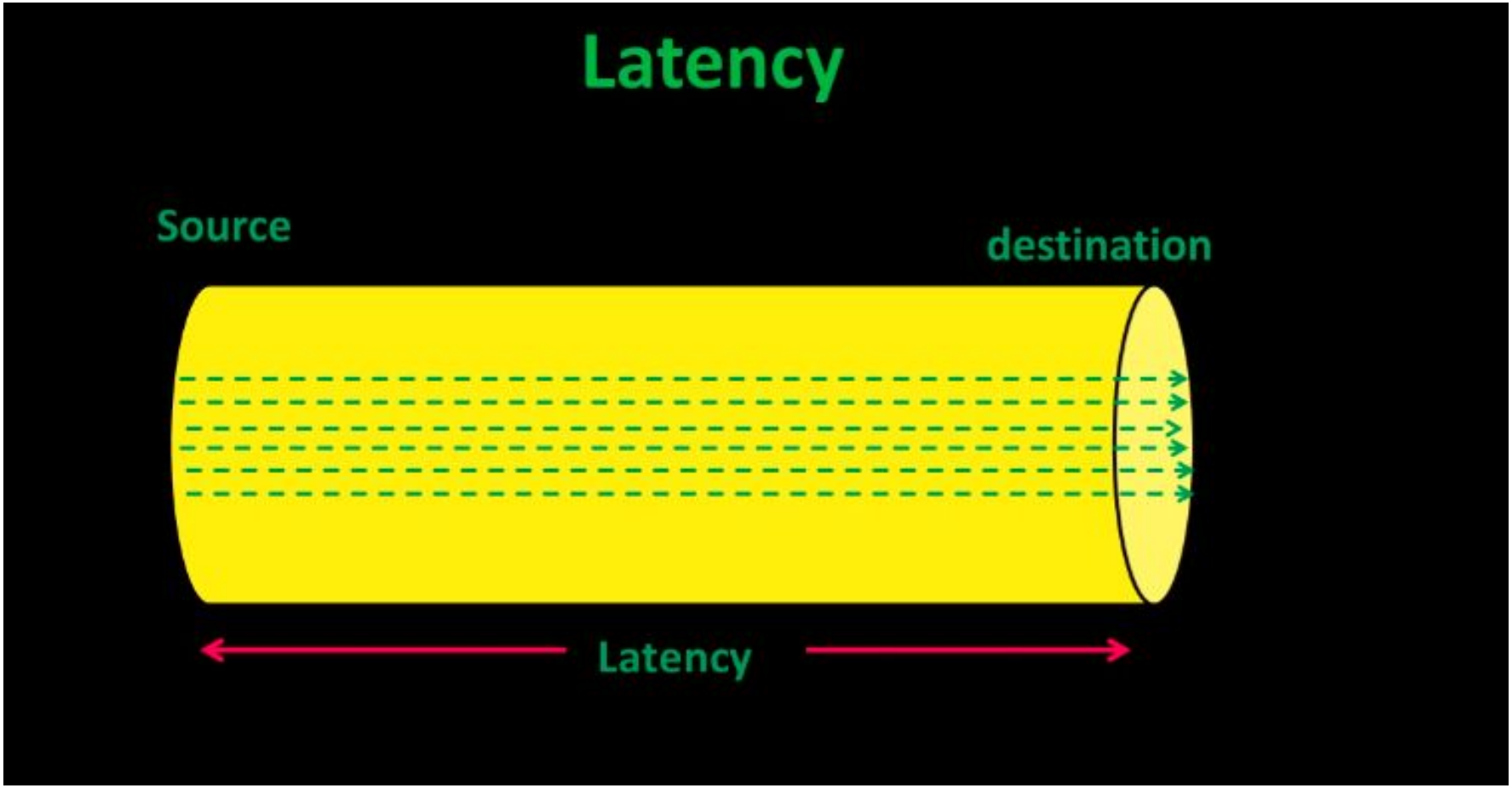
What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

Solution

We can calculate the propagation time as

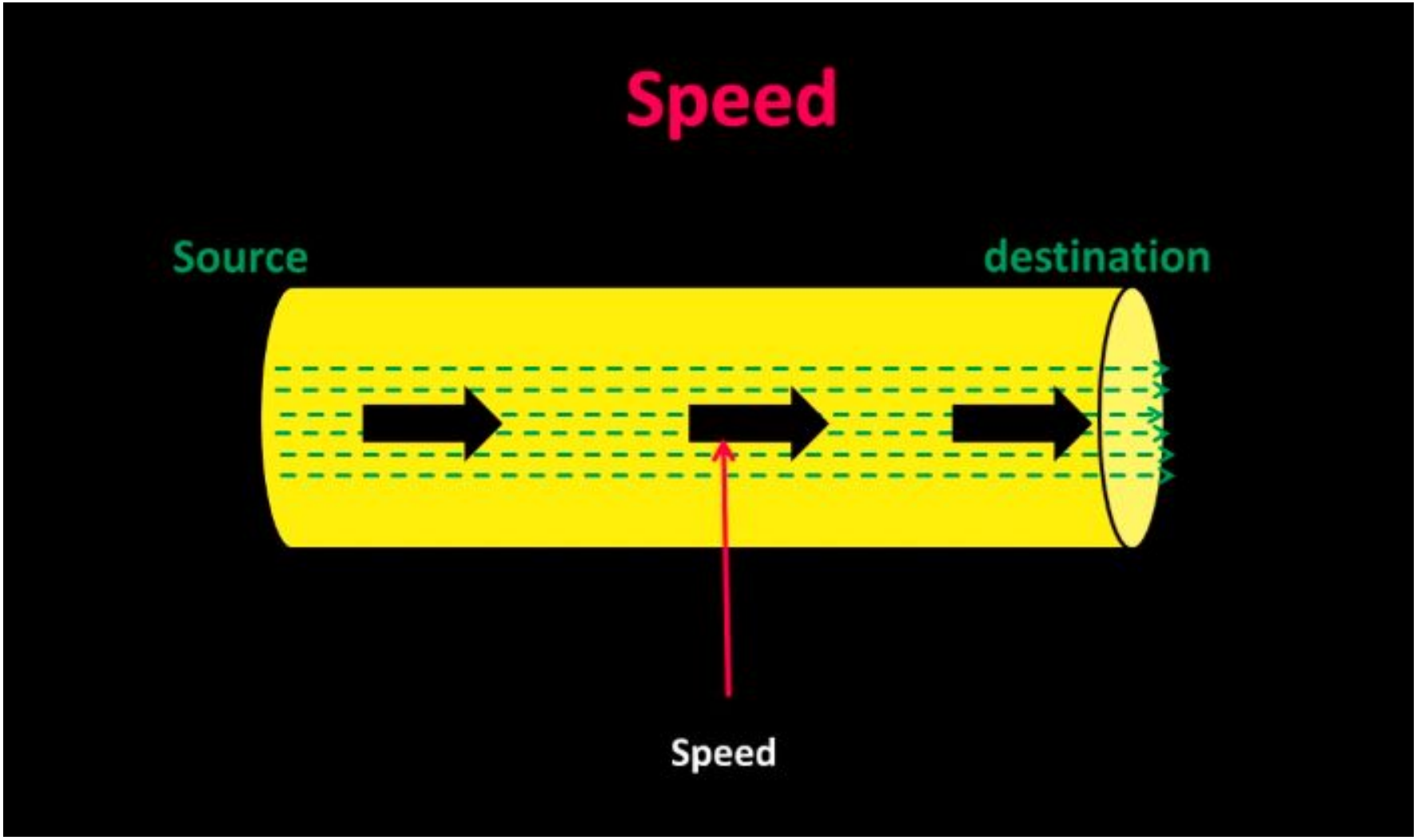
$$\text{Propagation time} = (12{,}000 \times 10{,}000) / (2.4 \times 2^8) = 50 \text{ ms}$$

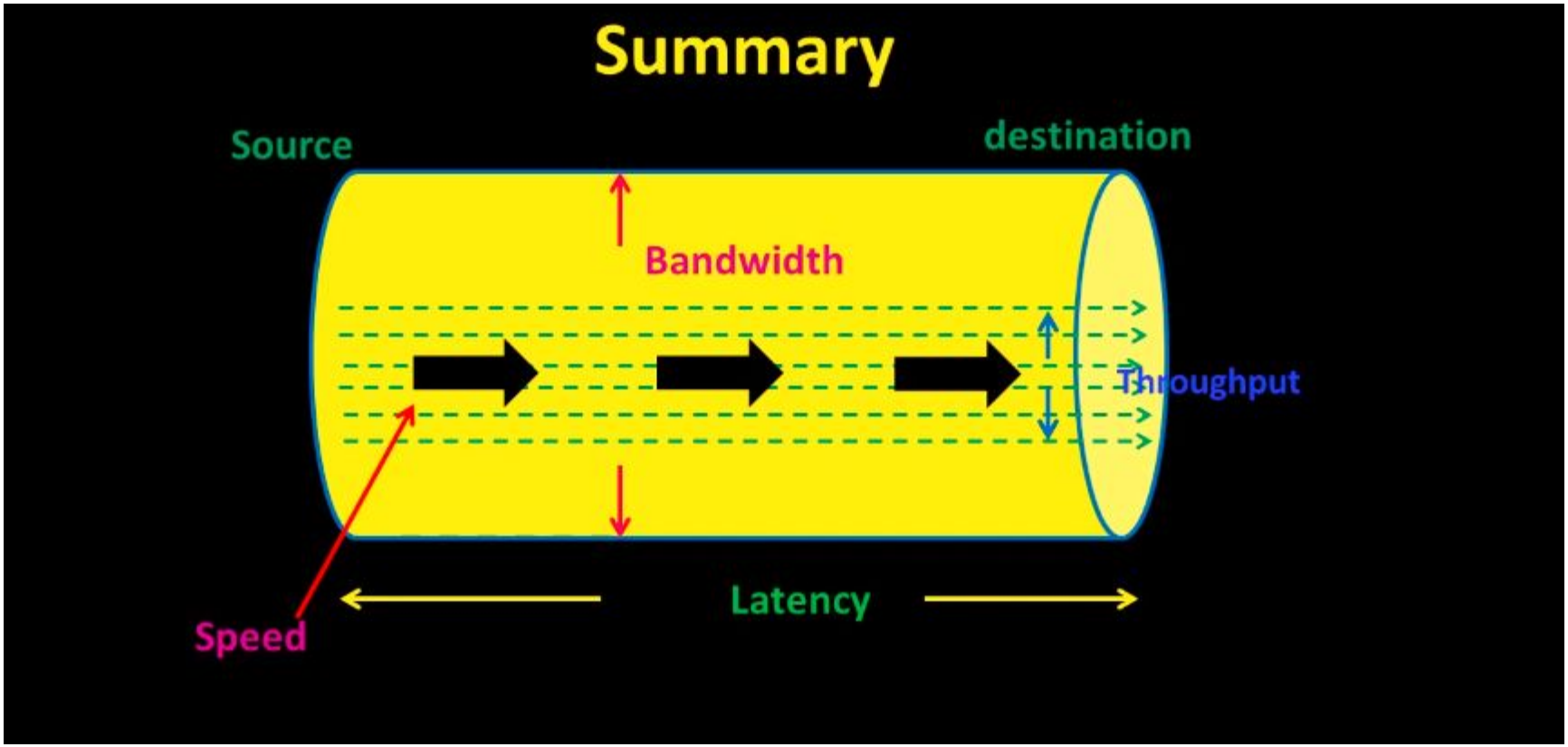
b


Latency
Source
destination
Latency

Speed

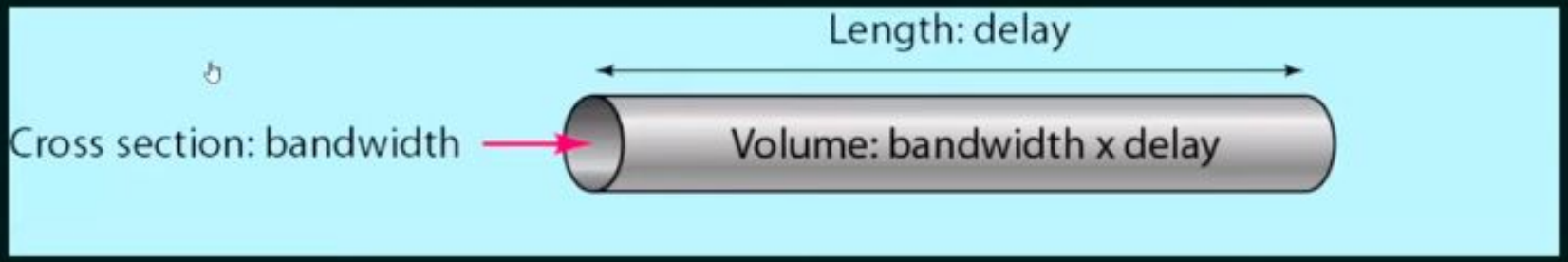
- Speed is **how fast that information** is received.
- Network speed measures **the transfer rate** in transmission.
- Speed is the bit rate of the circuit.
- Speed in measure in mbps, kbps, Mbps or MBps, kBps, GBps.


Speed
Source
destination
Speed

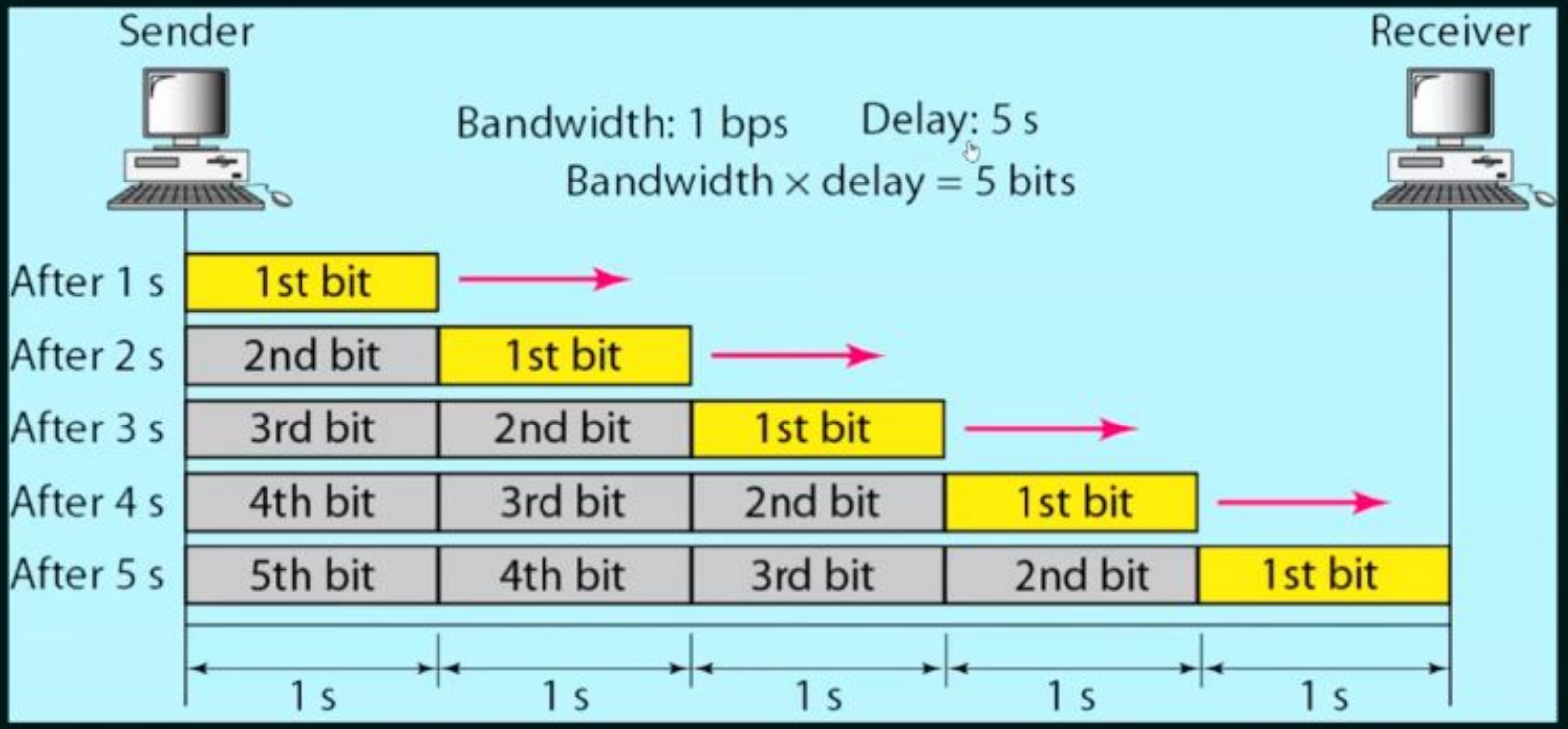

Summary
Source
destination
Bandwidth
Throughput
Latency
Speed

Bandwidth-Delay product

The bandwidth-delay product defines the number of bits that can fill the link.



Bandwidth-Delay product – Example



NESO ACADEMY

Bandwidth-Delay product – Solved Example

Consider that the link capacity of a channel is 512 Kbps and round – trip delay time is 1000ms.

Solution:

The bandwidth delay product = 512 Kbps × 1000 ms

= 512 × 1000 bits/sec × 1000 × 10^{-3} sec

= 512,000 bits

= 64,000 bytes

= 62.5 KB

ESO ACADEMY

Round Trip Time (RTT)

- ★ a.k.a Round Trip Delay Time (RTD).
- ★ It is the length of time it takes for a signal to be sent plus the length of time it takes for an acknowledgement of that signal to be received.